# Genetic Algorithm

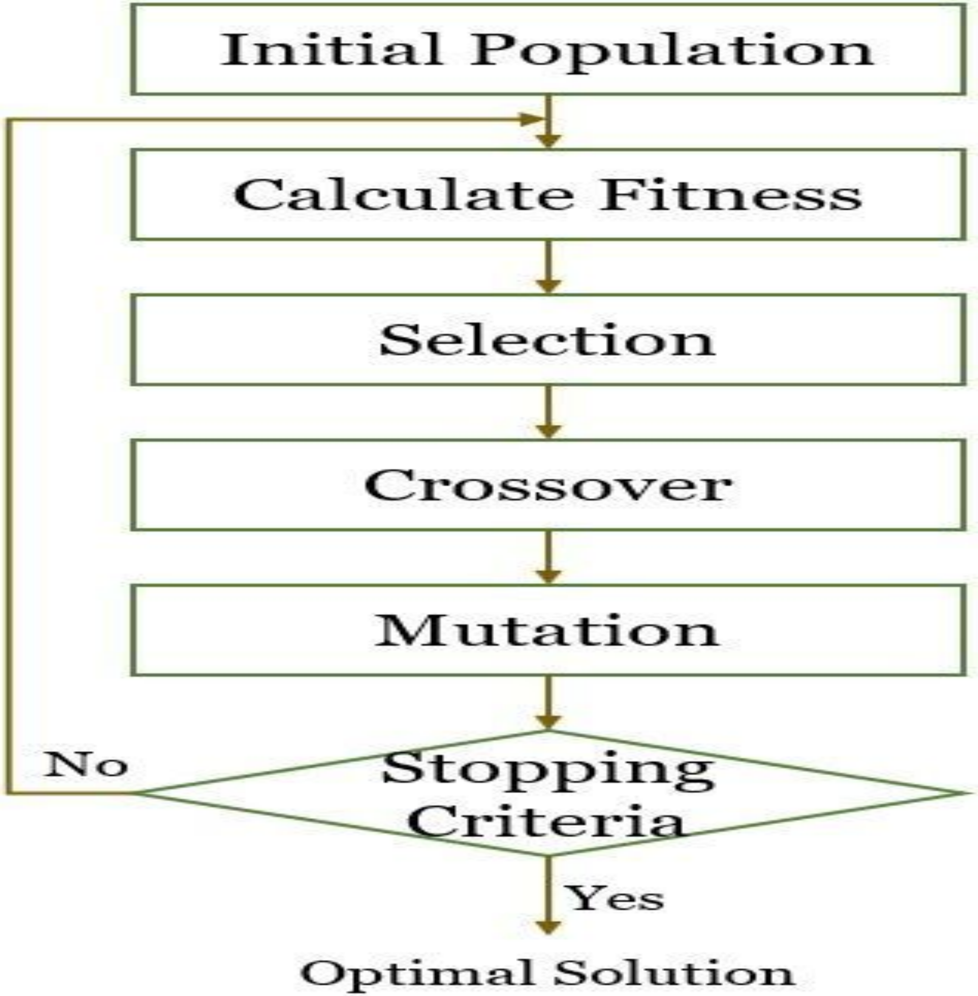
# INTRODUCTION TO GENETIC ALGORITHMS

- Simulating the logic of Darwinian selection, where only the best are selected for replication.
  - Over many generations, natural populations **evolve** according to the principles of natural selection and stated by Charles Darwin in The Origin of Species.
  - Only the **most suited elements** in a population are likely to survive and generate offspring, thus transmitting their biological heredity to new generations.
- Able to address complicated problems with many variables and a large number of possible outcomes by simulating the evolutionary process of **“survival of the fittest”** to reach a defined goal.
- Operate by **generating many random answers to a problem**, eliminating the worst and cross-pollinating better answers.
- **Repeating this elimination and regeneration process gradually improves the quality** of the answers to an optimal or near-optimal condition.

A **Genetic Algorithm (GA)** is a search and optimization technique in **Artificial Intelligence** inspired by the concept of **natural selection** (survival of the fittest). It is widely used to solve optimization problems where traditional methods struggle.

# Basic Idea of Genetic Algorithm

- We start with a **population** of possible solutions.
- Each solution is called a **chromosome**.
- Apply operations similar to biology:
    - **Selection** (choose best)
    - **Crossover** (combine solutions)
    - **Mutation** (random change)
- Repeat until we get the best solution.


Initial Population
Calculate Fitness
Selection
Crossover
Mutation
No
Stopping Criteria
Yes
Optimal Solution

## Steps of Genetic Algorithm

1. Initialize population
2. Evaluate fitness
3. Selection
4. Crossover
5. Mutation
6. Replacement
7. Repeat until goal reached

# Example : Maximize the function: f(x) = x^2

]

Where ( x ) is a **4-bit binary number** (0–15)

## ✅ Step 1: Initial Population

Let's randomly generate 4 chromosomes:

| **Chromosome (Binary)** | **Decimal (x)** |
|---|---|
| 1010 | 10 |
| 0110 | 6 |
| 1100 | 12 |
| 1001 | 9 |

## ✅ Step 2: Fitness Evaluation

$$f(x) = x^2$$

| Chromosome | x | Fitness = $x^2$ |
|---|---|---|
| 1010 | 10 | 100 |
| 0110 | 6 | 36 |
| 1100 | 12 | 144 |
| 1001 | 9 | 81 |

## ✅ Step 3: Selection (Roulette Wheel Idea)

Higher fitness → higher chance of selection

Total fitness = 100 + 36 + 144 + 81 = **361**

Probability example:

- 1100 → highest chance (144)
- 0110 → lowest chance (36)

👉 Assume selected parents:

- Parent 1: **1100**
- Parent 2: **1010**

## ✅ Step 4: Crossover

Choose crossover point (after 2 bits)

```
Parent 1: 11 | 00
Parent 2: 10 | 10
```

👉 Offspring:

- Child 1: **1110** (14)
- Child 2: **1000** (8)

---

## ✅ Step 5: Mutation

Randomly flip a bit (small probability)

Example:

- 1110 → 1111 (mutation at last bit)

## ✅ Step 6: New Population

Replace worst individuals:

| Chromosome | x | Fitness |
|---|---|---|
| 1111 | 15 | 225 |
| 1110 | 14 | 196 |
| 1100 | 12 | 144 |
| 1010 | 10 | 100 |

## ✅ Step 7: Check Termination

Best solution:

- **1111 (x = 15)**
- Fitness = **225 (maximum possible)**

✔️ Optimal solution found

---

## 🔷 Final Answer

👉 Maximum value of $f(x) = x^2$ occurs at:

$$x = 15, \quad f(x) = 225$$

Chromosome = solution

Gene = bit

Fitness function = objective function

Selection = survival of fittest

Crossover = recombination

Mutation = diversity